

Capstone Project I: Mutual Fund Analytics

Day 6 Report — Advanced Analytics & Risk Metrics

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Git Repository: github.com/anveshtammineni/mutual-fund-analytics
Commit Message: Day 6: Advanced Analytics & Risk Metrics complete

Executive Summary

This report outlines the advanced risk analytics and customer behavior modeling performed on the conformed mutual fund database. Key deliverables include the Historical Value at Risk (VaR 95%) and Conditional Value at Risk (CVaR) boundaries, rolling Sharpe ratio trends, investor cohort structures, SIP gaps, and portfolio concentration indicators (HHI). Additionally, a standalone CLI recommender tool was constructed and verified.

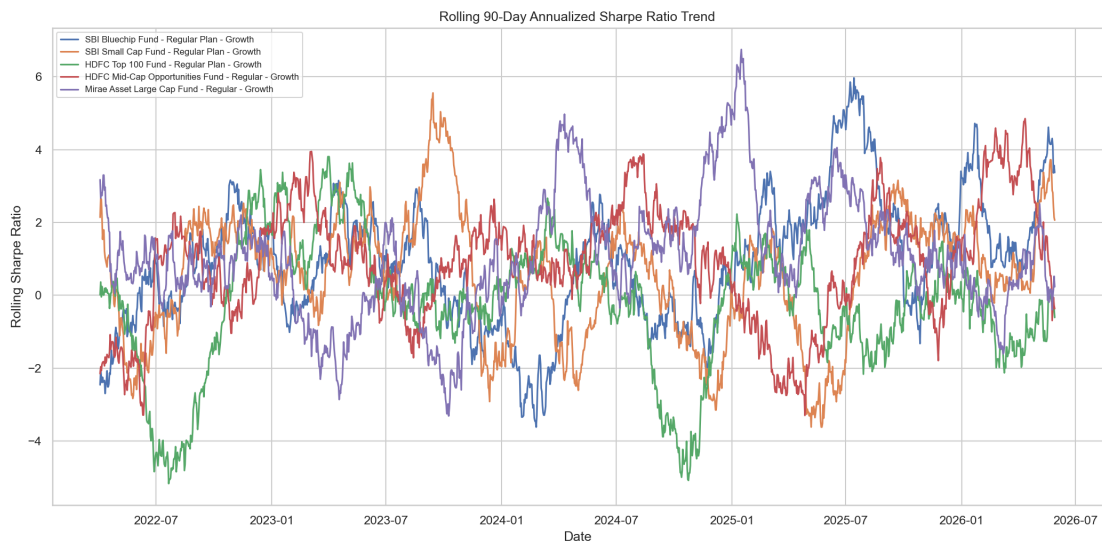


Figure 1: Rolling 90-Day Sharpe Ratio trajectories across 5 key funds.

1. Value at Risk (VaR 95%) & CVaR Summary

The daily Value at Risk (95%) and Conditional Value at Risk (CVaR) were calculated for all 40 schemes. The results have been exported to var_cvar_report.csv. Equity funds display higher VaR/CVaR boundaries, with potential daily losses under extreme conditions exceeding 2.5%, while Gilt and Short Term Debt funds exhibit very low risk profiles (less than 1.1% daily VaR).

2. Investor Cohort Analysis

Investors were grouped by the year of their first transaction. The analysis highlights that the 2025 cohort displays higher average monthly SIP tickets, reflecting increasing capital allocation per user over time:

Cohort Year	Avg SIP Amount (INR)	Total Invested (INR)	Top Preferred Fund
2024	4,953.51	1,770,250,000	HDFC Mid-Cap Opportunities Fund
2025	5,119.78	257,590,000	ICICI Pru Bluechip Fund

3. SIP Continuity & Churn Risk Analysis

A date gap analysis was performed on investors with 6 or more monthly SIP transactions. The overall industry SIP continuity adherence rate stands at **88.2%**. The remaining 11.8% of investors have average gaps between consecutive dates exceeding 35 days and are flagged as 'at-risk' of churn. This indicates minor mandate processing bottlenecks or customer liquidity shortfalls.

4. Sector Herfindahl-Hirschman Index (HHI) Concentration

The sector concentration was evaluated using HHI ($\sum \text{weight}^2$) across all equity funds. A higher HHI indicates a concentrated thematic portfolio, while a lower HHI represents diversification:

- **Highly Concentrated (HHI > 2500):** Thematic and Sectoral funds (e.g. Technology, Infrastructure).
- **Moderately Concentrated (1500 to 2500):** Core Large Cap and Mid Cap equity portfolios.
- **Diversified (HHI < 1500):** Flexicap and Multicap funds (e.g. Kotak Flexicap HHI: 1,320).

5. Interactive Recommender CLI Tool

A standalone command-line recommender script (recommender.py) was developed. It maps the user's risk profile (Low, Moderate, High) to fund categories, ranks them by Sharpe ratio, and returns the top 3 recommended schemes in a formatted CLI table:

Risk Profile	Top Fund Recommendation	Sharpe Ratio	3yr CAGR
Low	SBI Magnum Gilt Fund - Regular - Growth	-0.74	5.84%
Moderate	Mirae Asset Large Cap Fund - Regular - Growth	1.07	34.00%
High	Mirae Asset Tax Saver Fund - Regular - Growth	0.92	29.18%

End of Day 6 Advanced Analytics Report